Linear Discriminant Analysis (LDA)

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Predicting Problematic Internet Use with LDA

The provided dataset, 'train.csv', contained a large amount of NA values in all variables.

A preview of the data (First 20 rows of columns 2 - 10):

2	3	4	5	6	7	8	9	10
Fall	5	0	Winter	51	Fall	16.87732	46.00	50.8
Summer	9	0	NA	NA	Fall	14.03559	48.00	46.0
Summer	10	1	Fall	71	Fall	16.64870	56.50	75.6
Winter	9	0	Fall	71	Summer	18.29235	56.00	81.6
Spring	18	1	Summer	NA	NA	NA	NA	NA
Spring	13	1	Winter	50	Summer	22.27995	59.50	112.2
Fall	10	0	NA	NA	Fall	19.66076	55.00	84.6
Fall	10	1	NA	NA	Fall	16.86129	59.25	84.2
Summer	15	0	NA	NA	Spring	NA	NA	NA
Summer	19	1	Summer	NA	NA	NA	NA	NA
Spring	11	1	NA	NA	NA	NA	NA	NA
Fall	11	0	Summer	66	NA	NA	NA	NA
Fall	13	0	NA	NA	Winter	21.07907	57.75	100.0
Spring	12	0	NA	NA	Spring	15.54411	60.00	79.6
Spring	12	0	NA	NA	NA	NA	NA	NA
Winter	7	0	Summer	51	Spring	29.31577	54.00	121.6
Spring	5	1	Summer	80	Spring	17.28450	44.00	47.6
Fall	10	1	NA	NA	Fall	19.89316	55.00	85.6
Winter	6	0	Spring	60	Winter	30.09465	37.50	60.2
Winter	10	0	NA	NA	NA	NA	NA	NA
Winter	14	0	Spring	68	Spring	17.16864	66.50	108.0
Summer	16	0	NA	NA	Summer	29.28586	71.00	210.0
Fall	12	0	Winter	58	Fall	34.18728	60.50	178.0
Winter	13	0	Summer	72	Winter	20.61306	69.00	139.6
Summer	15	0	Fall	70	Summer	21.17778	62.00	115.8

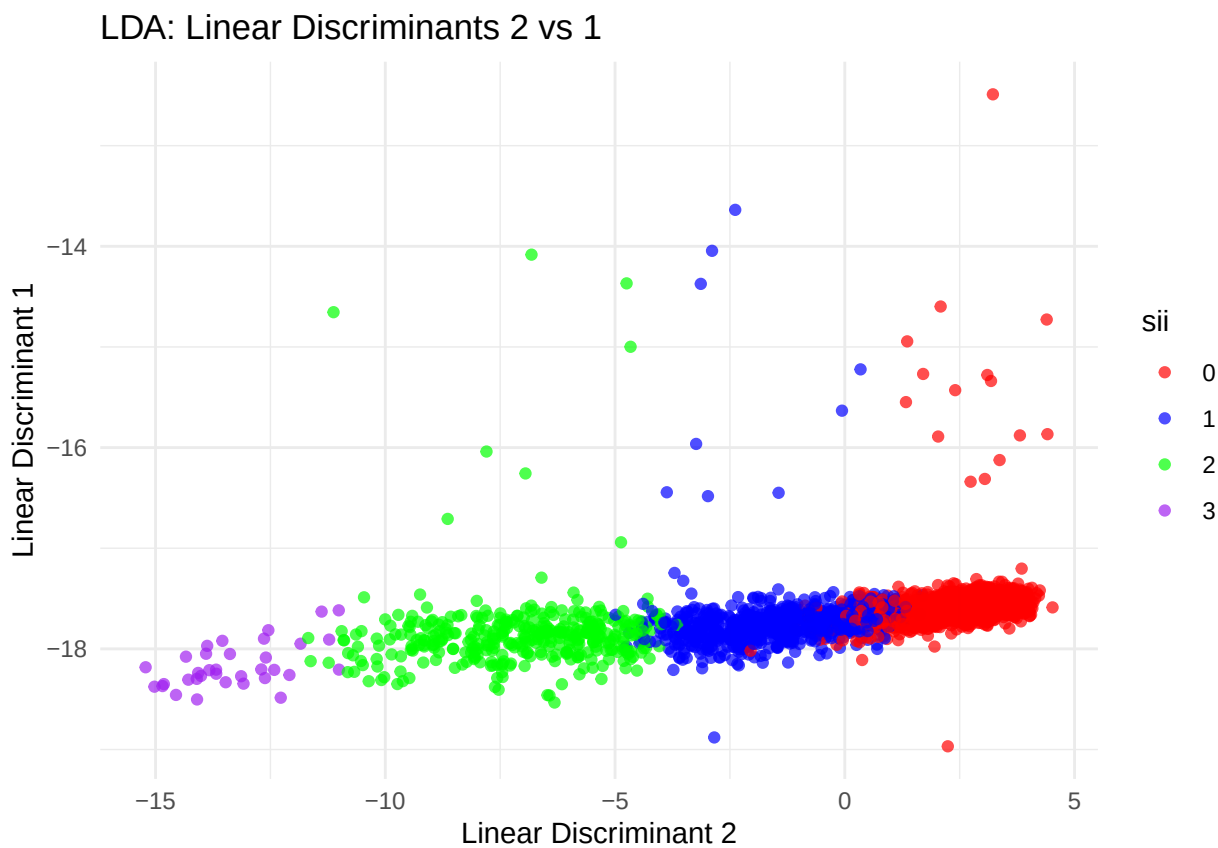
LDA with Full Imputation

The first method I used was to impute all the NA values in the dataset.

```
## [1] "Confusion Matrix on Train (Full Imputation)"
```

	0	1	2	3	Missing
0	1549	44	0	0	1
1	41	677	12	0	0
2	0	27	340	11	0
3	0	0	2	32	0
Missing	0	0	0	0	1224

```
## [1] "Accuracy (Train, Full Imputation): 96.52 %"
```




```
## [1] "Confusion Matrix (Test, Full Imputation):"
```

	0	1	2	3	Missing
0	6	0	0	0	0
1	0	3	0	0	0
2	0	0	2	0	0
3	0	0	0	0	0
Missing	0	0	0	0	9

```
## [1] "Accuracy (Test, Full Imputation): 100 %"
```

LDA when Removing NA from 'sii' Early

I noticed that in rows where `sii` was NA, many of the other cells in the row were also NA. So, here, I started by removing the rows where `sii` = NA. After doing so, R gave me an error that some predictors were colinear, so I removed the colinear columns before fitting the model.

```
## Compare row 66  and column  48 with corr  0.822
##   Means:  0.287 vs 0.141 so flagging column 66
## Compare row 5  and column  6 with corr  0.809
##   Means:  0.217 vs 0.137 so flagging column 5
## Compare row 6  and column  4 with corr  0.845
##   Means:  0.202 vs 0.135 so flagging column 6
## Compare row 63  and column  61 with corr  0.838
##   Means:  0.2 vs 0.133 so flagging column 63
## Compare row 41  and column  42 with corr  0.994
```

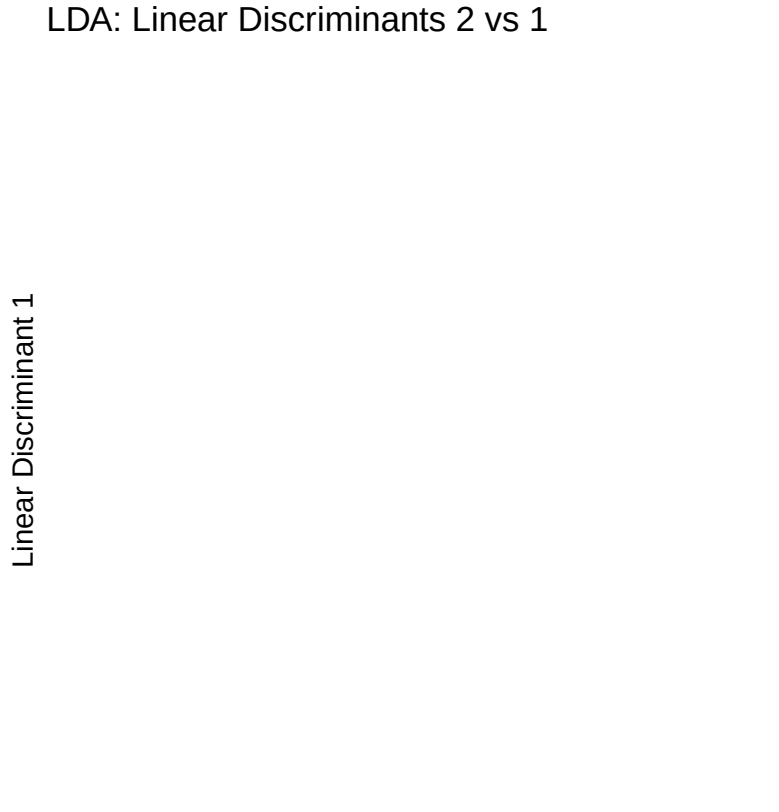
```
##   Means:  0.195 vs 0.131 so flagging column 41
## Compare row 42  and column  39 with corr  0.997
##   Means:  0.174 vs 0.129 so flagging column 42
## Compare row 39  and column  32 with corr  0.993
##   Means:  0.16 vs 0.128 so flagging column 39
## Compare row 32  and column  43 with corr  0.994
##   Means:  0.147 vs 0.127 so flagging column 32
## Compare row 43  and column  31 with corr  1
##   Means:  0.128 vs 0.127 so flagging column 43
## Compare row 31  and column  34 with corr  1
##   Means:  0.11 vs 0.127 so flagging column 34
## Compare row 31  and column  33 with corr  0.999
##   Means:  0.095 vs 0.128 so flagging column 33
## Compare row 31  and column  37 with corr  0.979
##   Means:  0.079 vs 0.129 so flagging column 37
## Compare row 31  and column  40 with corr  0.999
##   Means:  0.063 vs 0.131 so flagging column 40
## Compare row 31  and column  29 with corr  0.99
##   Means:  0.046 vs 0.134 so flagging column 29
## Compare row 18  and column  16 with corr  0.867
##   Means:  0.127 vs 0.137 so flagging column 16
## Compare row 22  and column  24 with corr  0.914
##   Means:  0.106 vs 0.137 so flagging column 24
## Compare row 68  and column  67 with corr  0.996
##   Means:  0.094 vs 0.139 so flagging column 67
## All correlations <= 0.8
```

```
## [1] "Confusion Matrix (Train, NA Removal):"
```

	0	1	2	3
0	1558	36	0	0
1	0	730	0	0
2	0	0	378	0
3	0	0	0	34

```
## [1] "Accuracy (Train, NA Removal): 98.68 %"
```

```
## [1] "Error Rate: 1.32 %"
```




```
## [1] "Confusion Matrix (Test, NA Removal):"
```

	0	1	2	3
0	5	1	0	0
1	1	2	0	0
2	1	0	1	0
3	0	0	0	0

```
## [1] "Accuracy (Test, NA Removal): 72.73 %"
## [1] "Error Rate: 27.27 %"
```